

An Implementation of Attribution-By-Design for Ethical AI Music

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Background

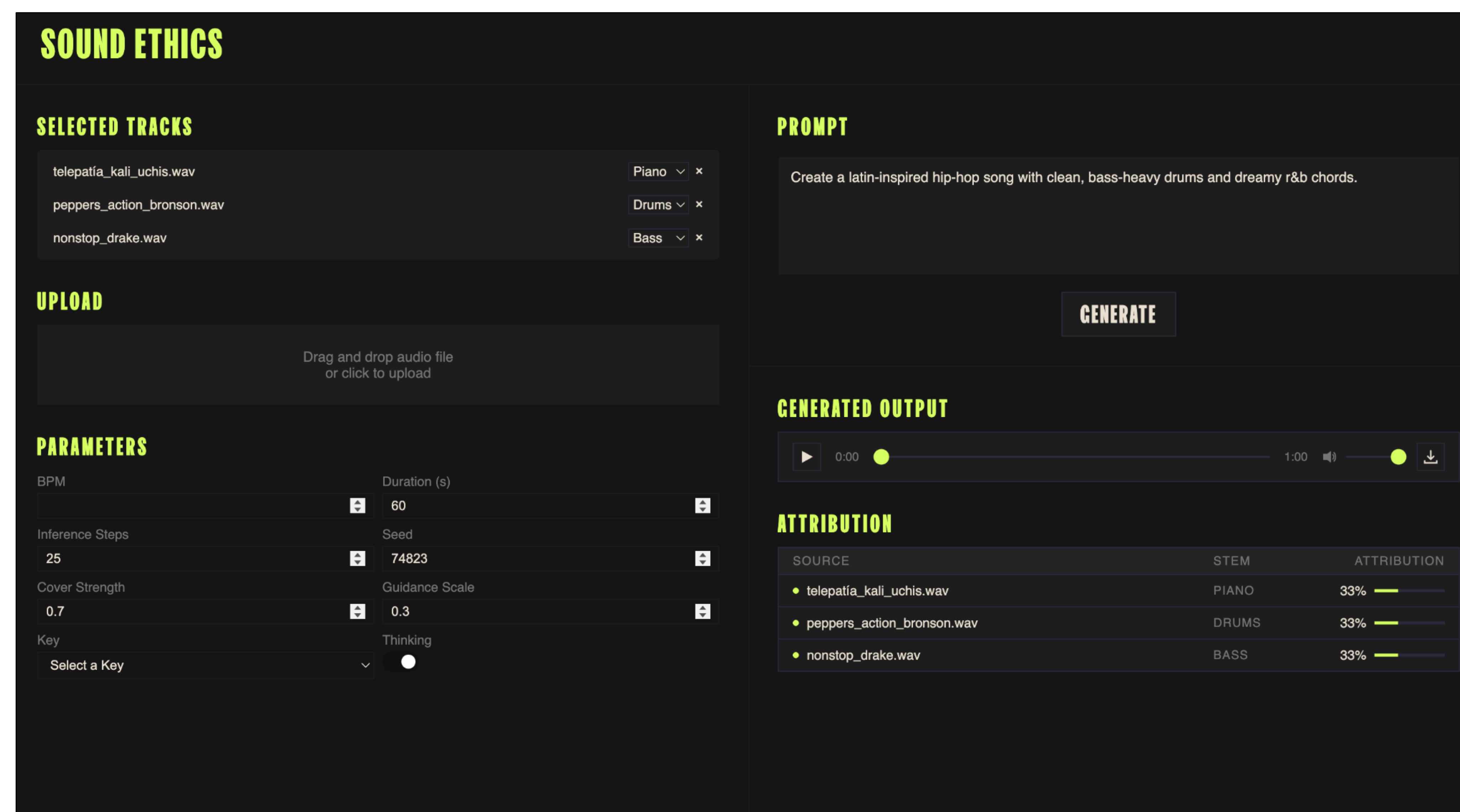
Sound Ethics is an organization that develops and advocates for ethical Artificial Intelligence (AI). Their aim is to ensure that musicians and artists are able to thrive in music careers alongside the development of generative music models [1]. We have partnered with Sound Ethics to implement an AI framework that generates music and gives proper attribution to the original artists. This implementation, based on the Sony AI's **Attribution-by-Design** architecture, can be used to ethically generate new music and enables developers to give proper credit and compensation to the original artists [2].

User Interface

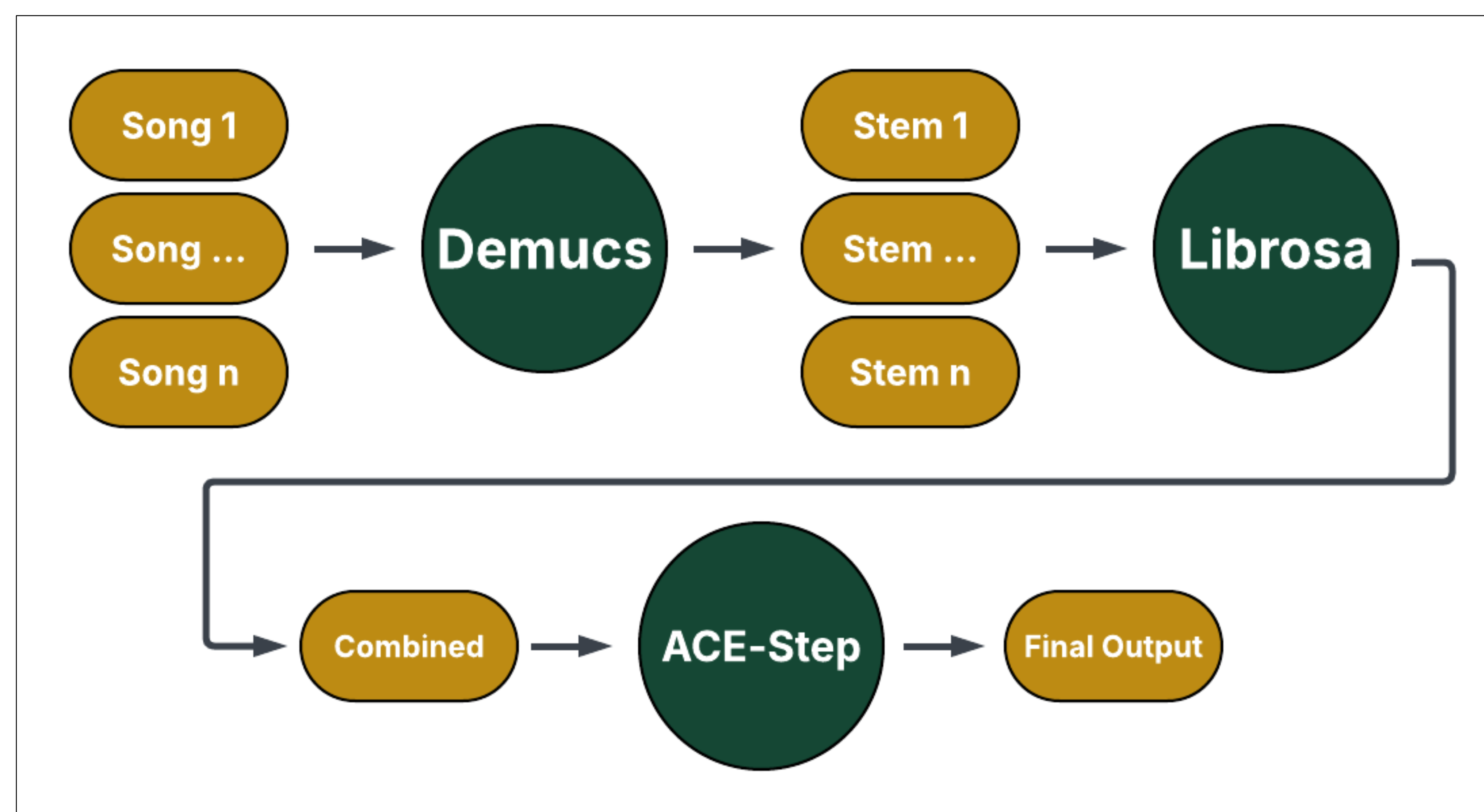
- The user can upload audio clips, provide a text prompt, and specify various output features, including key, tempo, and duration.
- We **emphasize creative control** for two reasons:
 - To produce outputs that match user intent.
 - To align with the Sony AI Team's definition of attribution, where explicit user choices are a requirement for proper attribution.

Backend Overview

- Tools:
 - Demucs** isolates specific audio components from the original audio (ie. drums)
 - Librosa** combines multiple audio files
 - ACE-Step** generates music conditioned on a text description and a single audio input [3]
- Users can condition on up to **5 audio components**: drums, guitar, piano, bass, melody (other).



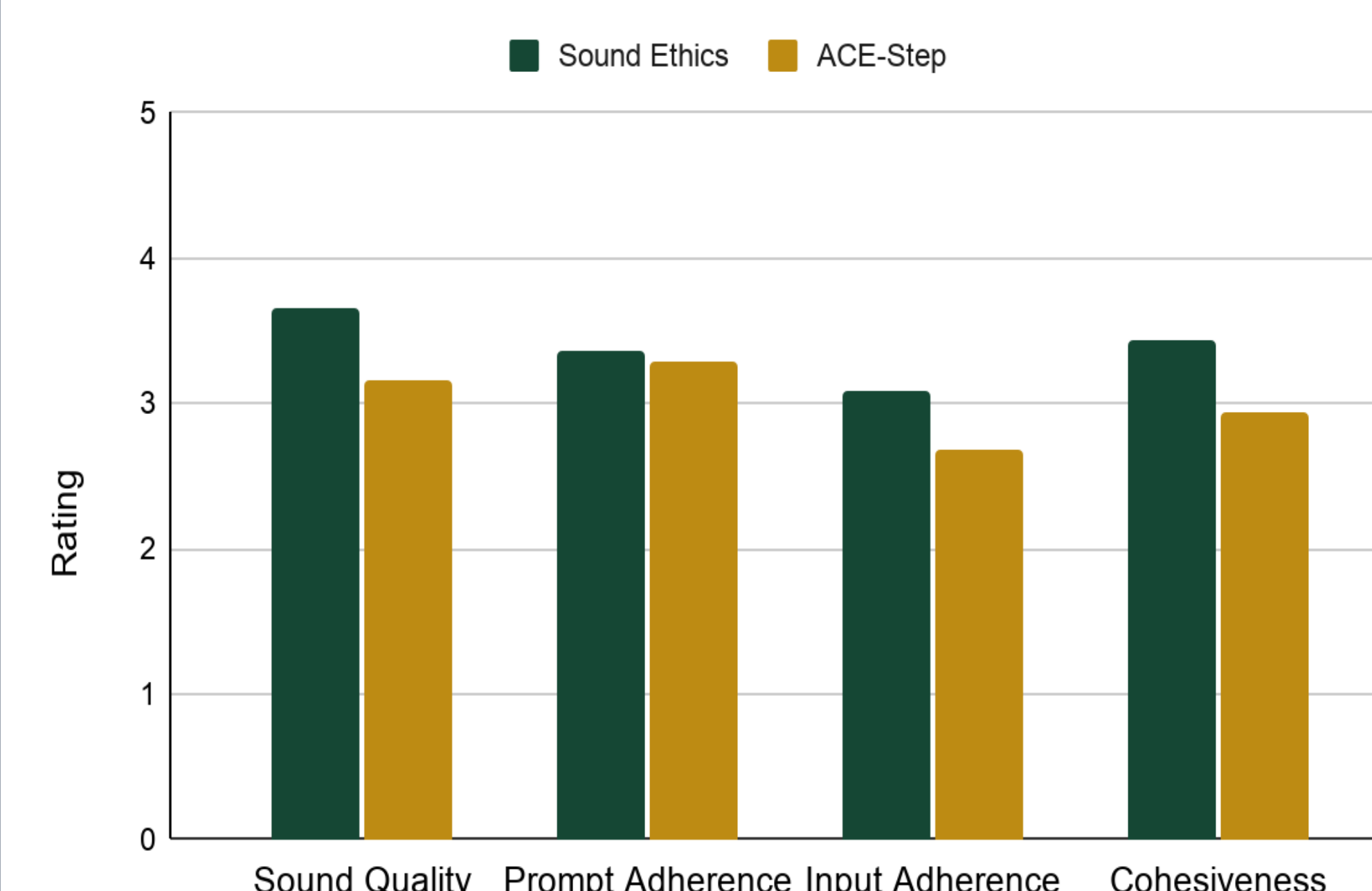
The user interface gives users maximum control over music generation. Users can upload multiple audio clips that are used to condition the output.



The backend combines existing technologies to create a new workflow for music generation. This workflow allows for conditioning on multiple songs without needing to create or train a new model.

Survey Results

Comparing Output Quality Between Frameworks
14 Survey Responses



- Our **survey assesses**: sound quality, prompt adherence, audio adherence, and overall audio cohesiveness

Conclusions

We have implemented a music generation framework that **emphasises creative control** and **gives proper attribution** to reference audio. This framework aligns with the Sony AI Team's **Attribution-by-Design framework**, in which explicit user choices create opportunities for proper attribution. Our framework **outperforms base ACE-Step** across all metrics. Most importantly, our input adherence is substantially higher than base ACE-Step, which means that our framework's outputs **more strongly align with input audios**.

References

- [1] Sound Ethics. <https://www.soundethics.org/>
- [2] Fabio Morreale, Wiebke Hutiri, Joan Serrà, Alice Xiang, and Yuki Mitsufuji. (2025). *Attribution-By-Design: Ensuring Inference-Time Provenance in Generative Music Systems*. arXiv. <https://doi.org/10.48550/arXiv.2510.08062>
- [3] Gong, Junmin, et al. "Ace-step: A step towards music generation foundation model." arXiv preprint arXiv:2506.00045 (2025).